EXPERIMENT-1

Basic Networking Commands in Linux and Windows operating systems

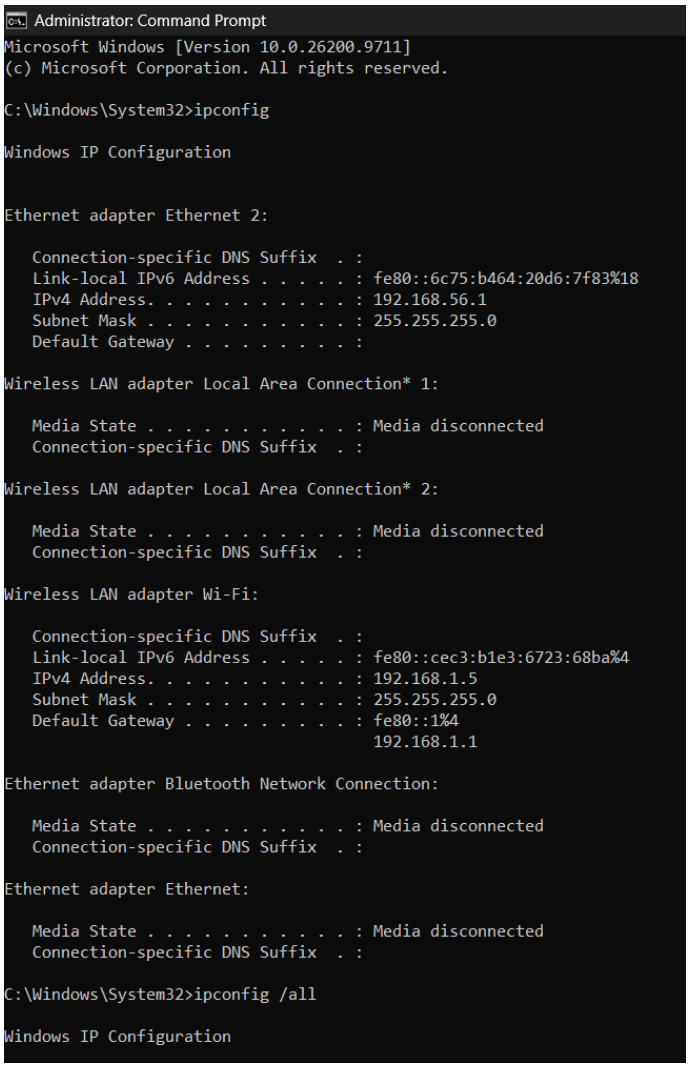
Aim: To study and execute basic networking commands in Windows and Linux operating systems for viewing network configuration, testing connectivity, checking routing information, monitoring network status, and troubleshooting network-related issues.

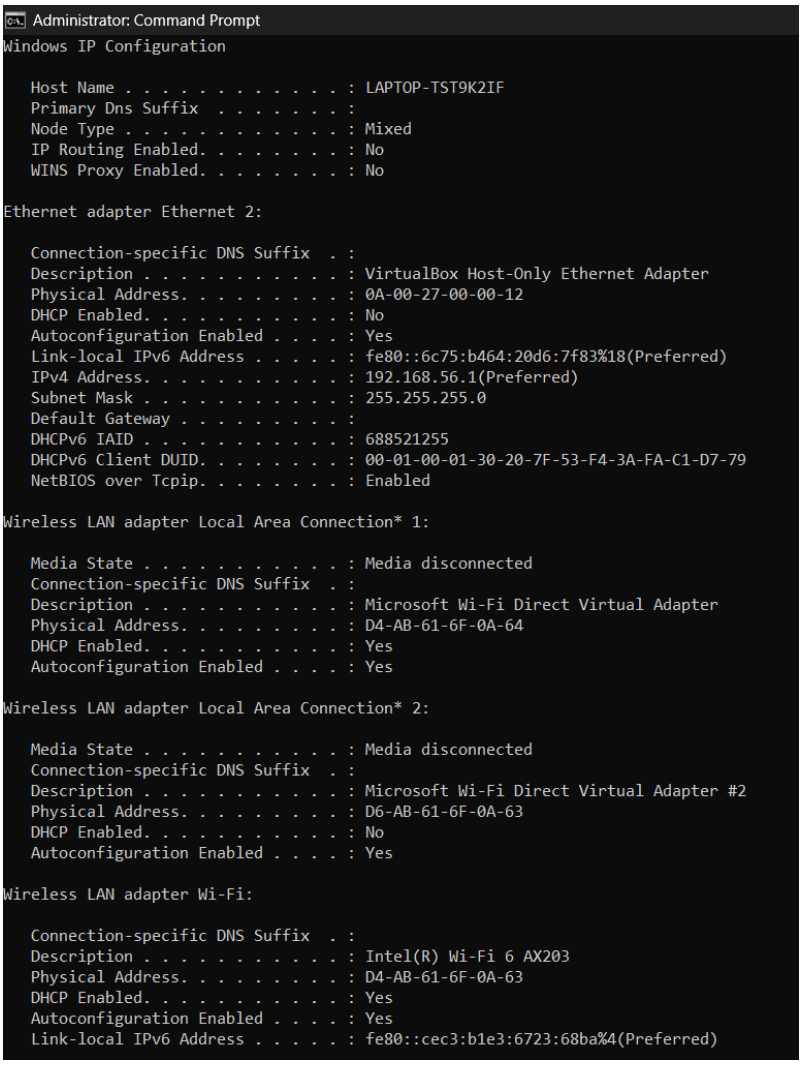
Software Requirements :

• Windows 10/11 or Linux (Ubuntu 20.04/22.04)

• Command Prompt (Windows)

• Terminal (Linux)





```
C:\Windows\System32>ping google.com

Pinging google.com [142.250.205.78] with 32 bytes of data:
Reply from 142.250.205.78: bytes=32 time=5ms TTL=119
Reply from 142.250.205.78: bytes=32 time=5ms TTL=119
Reply from 142.250.205.78: bytes=32 time=6ms TTL=119
Reply from 142.250.205.78: bytes=32 time=6ms TTL=119

Ping statistics for 142.250.205.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 6ms, Average = 5ms

C:\Windows\System32>tracert google.com

Tracing route to google.com [142.250.205.78]
over a maximum of 30 hops:

  1     3 ms     1 ms     1 ms  192.168.1.1
  2     8 ms    11 ms     5 ms  110.4.27.73.hathway.com [27.4.110.73]
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    80 ms     9 ms     8 ms  pnmaaa-ar-in-f14.1e100.net [142.250.205.78]

Trace complete.

C:\Windows\System32>nslookup google.com
Server:  UnKnown
Address:  192.168.1.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:832::200e
          192.178.211.100
          192.178.211.102
          192.178.211.138
          192.178.211.113
          192.178.211.101
          192.178.211.139


C:\Windows\System32>route print
===========================================================================
Interface List
 18...0a 00 27 00 00 12 ......VirtualBox Host-Only Ethernet Adapter
  3...d4 ab 61 6f 0a 64 ......Microsoft Wi-Fi Direct Virtual Adapter
 10...d6 ab 61 6f 0a 63 ......Microsoft Wi-Fi Direct Virtual Adapter #2
 11...f4 3a fa c1 d7 79 ......Realtek PCIe GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.1.1      192.168.1.5     50
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      192.168.1.0    255.255.255.0         On-link       192.168.1.5    306
      192.168.1.5  255.255.255.255         On-link       192.168.1.5    306
    192.168.1.255  255.255.255.255         On-link       192.168.1.5    306
     192.168.56.0    255.255.255.0         On-link      192.168.56.1    281
     192.168.56.1  255.255.255.255         On-link      192.168.56.1    281
   192.168.56.255  255.255.255.255         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link       192.168.1.5    306
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link      192.168.56.1    281
  255.255.255.255  255.255.255.255         On-link       192.168.1.5    306
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  4    306 ::/0                     fe80::1
  1    331 ::1/128                  On-link
 18    281 fe80::/64                On-link
  4    306 fe80::/64                On-link
 18    281 fe80::6c75:b464:20d6:7f83/128
                                    On-link
  4    306 fe80::cec3:b1e3:6723:68ba/128
                                    On-link
  1    331 ff00::/8                 On-link
 18    281 ff00::/8                 On-link
  4    306 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None

C:\Windows\System32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49674        LAPTOP-TST9K2IF:49675  ESTABLISHED
  TCP    127.0.0.1:49675        LAPTOP-TST9K2IF:49674  ESTABLISHED
  TCP    192.168.1.5:50338      192.168.1.4:8009       ESTABLISHED
  TCP    192.168.1.5:52141      104.18.36.252:https    ESTABLISHED
  TCP    192.168.1.5:52522      lcbomp-in-f188:5228    ESTABLISHED
  TCP    192.168.1.5:52912      1:https                TIME_WAIT
  TCP    192.168.1.5:52920      40.104.146.162:https   TIME_WAIT
  TCP    192.168.1.5:52927      static:http            TIME_WAIT
  TCP    192.168.1.5:53021      sa-in-f188:5228        ESTABLISHED
  TCP    192.168.1.5:53155      172.67.186.25:https    TIME_WAIT
  TCP    192.168.1.5:53188      104.21.76.26:https     ESTABLISHED
  TCP    192.168.1.5:54383      172.64.150.28:https    TIME_WAIT
  TCP    192.168.1.5:54495      52.98.56.210:https     TIME_WAIT
  TCP    192.168.1.5:55270      192.168.1.4:8009       ESTABLISHED
  TCP    192.168.1.5:55418      whatsapp-cdn-shv-02-maa3:https  ESTABLISHED
  TCP    192.168.1.5:55604      lcbomp-in-f188:5228    ESTABLISHED
  TCP    192.168.1.5:56644      4.213.25.240:https     ESTABLISHED
  TCP    192.168.1.5:57223      sn-in-f188:5228        ESTABLISHED
  TCP    192.168.1.5:57344      172.64.155.209:https   TIME_WAIT
  TCP    192.168.1.5:60571      172.64.155.209:https   ESTABLISHED
  TCP    192.168.1.5:63055      172.64.145.54:https    CLOSE_WAIT
  TCP    192.168.1.5:63058      104.18.36.61:https     CLOSE_WAIT
  TCP    192.168.1.5:63616      104.18.37.228:https    ESTABLISHED
  TCP    192.168.1.5:63617      20.42.65.84:https      ESTABLISHED
  TCP    192.168.1.5:63618      52.123.129.14:https    ESTABLISHED
  TCP    192.168.1.5:63997      172.188.155.25:https   ESTABLISHED

C:\Windows\System32>netstat -a

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:445            LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:3306           LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:5040           LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:33060          LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:49664          LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:49665          LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:49666          LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:49667          LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:49668          LAPTOP-TST9K2IF:0      LISTENING
  TCP    0.0.0.0:49677          LAPTOP-TST9K2IF:0      LISTENING

C:\Windows\System32>getmac

Physical Address    Transport Name
=================== ==========================================================
F4-3A-FA-C1-D7-79   Media disconnected
D4-AB-61-6F-0A-67   Media disconnected
D4-AB-61-6F-0A-63   \Device\Tcpip_{1EA7D478-B256-4675-96D7-2F7B0F11B905}
0A-00-27-00-00-12   \Device\Tcpip_{F4A63484-1407-4F78-A027-5481FC323255}

C:\Windows\System32>hostname
LAPTOP-TST9K2IF

C:\Windows\System32>arp -a

Interface: 192.168.1.5 --- 0x4
  Internet Address      Physical Address      Type
  192.168.1.1           44-95-3b-e8-52-40     dynamic
  192.168.1.4           40-49-0f-24-46-5f     dynamic
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x12
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

C:\Windows\System32>
```